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Powering the brave new blockchain world

Every small shift today comes with the promise of a revolutionary change that has the potential to transform our daily lives. These transformations, though subjected to initial scepticism, pave way for a renewed ecosystem for people. We may be living through a similar moment today as more people across multiple industries highlight the potential for blockchain technologies to create a similar shift.

Blockchain technology is a way to record data and, increasingly, execute complex business processes like smart contracts in a cryptographical secured environment without using a middleman or institution in-between. It consists of a sequential, ever-growing, timestamped set of records that are grouped in blocks and maintained by many participants. So instead of a bank owning the ledger with your account balance, each customer has a role in maintaining a decentralized ledger. No one party controls the system—it is truly “peer-to-peer”—and the data exists in multiple places. Each block is interdependent and making alterations of records is uneconomical, if not impossible. Any changes to the block are tracked and must be agreed to by the majority of the blockchain. This process creates a more secure system of transactions by ensuring the provenance of any change is accurate and traceable.

While the roots of blockchain technology can be found in the world

of cryptocurrencies, many organizations are exploring how they can harness blockchains to transform their businesses, enable more efficient markets, or create new products and services. In financial services, there is potential to reduce trade settlement times from days to seconds, streamline audit processes, and speed up trade finance negotiation. More than that, blockchain technology can be used to help ensure conflict-free diamonds, protect against counterfeiting using IoT, and reliably track the provenance on an item from source to delivery. Companies are also starting to apply blockchain to anti-counterfeit, supply chain, and health records. Already, companies including Overstock and Paypal are accepting payments that are made on blockchain-based currencies.

Scalability is a major ceiling for current blockchain implementations, making it difficult for the technology to be applied at scale to applications like payments. For instance, VISA processes 1,667 transactions every second. Bitcoin has a capacity of 3-4 transactions per second, and Ethereum supports up to 15 per second. Even more significantly, blockchains today require colossal amounts of electricity.

To provide some context, it is estimated that for the Bitcoin network to process 3 to 4 transactions per second it requires 32 terawatt hours of electricity annually — about as much

as consumed by the country of Denmark. Scaling this to handle a credit card network isn't possible from either a technical or energy perspective.

To enable blockchain and other next generation technologies such as artificial intelligence, and augmented and virtual reality, the semiconductor industry must find new ways to build processors that deliver more performance and operate more efficiently. From 3D chip stacking to better circuitry design, engineers are tackling these challenges. Tremendous advances are being made in this area by processors including CPUs and GPUs. These advances are achieved through architecture innovation, improved power management, and new packaging options. Standardizing blockchains on these broadly available processors increases the potential to attract users and further focuses research and development on improving performance and efficiency.

It is early days for blockchain, but the prospects are intriguing. In fact, industry expert Don Tapscott believes this is a true paradigm shift. His view is that blockchain could revolutionize the world economy by levelling the playing field regardless of participant economic status and create new opportunities for value creation.

Immutable, distributed ledgers and smart contracts that improve finance are only the beginning. For example, blockchain has the potential to enable secure lifetime medical record sharing across providers, overcoming the differences in implementation of electronic medical records and improving healthcare for all. As a time-stamped, programmable ledger, blockchain could track and correlate all the health records for a person, and present a chronological and complete picture of care at any point in time.

It is a brave new world that can only be realized if we solve the performance, efficiency, and cost challenges to power blockchains. Through smart engineering and a true commitment to energy efficiency, the semiconductor industry can play an important role in supporting the growth of blockchain and other advanced technologies. **d**